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BIOMEDICAL LIMITED

pH METER QUALIFICATION PROTOCOL

Installation Qualification / Operational Qualification / Performance Qualification

LOCATION – WTP/Lab
PROTOCOL No. – BML/QA/pHQ/001
Effective date: August, 2025.

PROVIDED BY:
BIOMEDICAL LIMTED
KWARA STATE, NIGERIA

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PROTOCOL FORMAT APPROVAL:

Signatures on this page indicate that this document and its format have been prepared, checked and approved by the personnel concerned. It shall be signed and date as shown below

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Signature:		
Position:		
Date:		

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Position:		
Date:		

1. Purpose

This protocol defines the Installation Qualification (IQ), Operational Qualification (OQ), and Performance Qualification (PQ) requirements for the pH meter designated for use in the LVP Plant and Quality Control laboratory at Biomedical Ltd (BML). The document establishes documented evidence that the instrument is correctly installed, operates within defined parameters, and consistently performs accurate pH measurements suitable for GMP-regulated pharmaceutical manufacturing.

Successful completion of all three qualification phases, as documented in the companion Qualification Report (BML/EQ-RPT/PHM/001), is required before the instrument is approved for routine use.

2. Scope

This protocol applies to the pH meter installed for in-process pH measurement, final product release testing, and cleaning validation support within BML's LVP Plant. It covers:

- The main pH meter unit and all integrated accessories (electrode(s), temperature probe, ATC module, calibration solutions)
- Any spare or replacement electrode of the same specification
- pH measurements performed on LVP products: Normal Saline 0.9%, Dextrose 5%, Ringer's Lactate, and CAPD solutions

This protocol does not cover the Oral Liquid Plant pH meter, which is qualified under a separate document.

3. Regulatory and Reference Basis

Reference	Applicability
EU GMP Annex 15 (2015)	Qualification and Validation framework
EU GMP Chapter 4	Documentation requirements
ICH Q2(R1)	Analytical method validation – applies to instrument performance
USP <791> pH	pH measurement requirements for pharmaceutical products
Ph.Eur. 2.2.3	Potentiometric determination of pH
WHO TRS 937 Annex 4	WHO GMP for pharmaceutical products
NAFDAC GMP Guidelines	National regulatory requirements (Nigeria)
BML/SOP/QC/005	Operation and Calibration of pH Meter
BML/CAL/REG/001	BML Calibration Register and Schedule

4. Instrument Description and Identification

Parameter	Details
Equipment Name	pH Meter – Laboratory/Process Grade
Manufacturer	[To be completed on receipt of equipment]
Model	[To be completed]
Serial Number	[To be completed]
Electrode Type	[e.g., Combined glass-reference, ATC-capable]
Measurement Range	pH 0.00 – 14.00
Resolution	0.01 pH units
Accuracy Specification	±0.01 pH units (manufacturer's specification)
Temperature Compensation	Automatic (ATC), 0 – 100°C
Power Supply	[Mains / Battery – to be confirmed]
Location	LVP Plant – QC Laboratory / WTP
Calibration Interval	Daily (before use) / External: 3 months

5. Roles and Responsibilities

Role	Responsibility
QA officer	Protocol authorship, test execution oversight, review of results, preparation of the Qualification Report
QA Manager	Review and approval of protocol and report; final release decision
Production Manager	Provision of installation location; operational approval
Plant Engineer / Maintenance	Physical installation verification, electrical safety checks
QC Analyst	Execution of OQ and PQ tests under the supervision of QAQC Head

6. Pre-Qualification Requirements

The following items shall be confirmed before commencing any qualification activity:

Pre-Qualification Check	Pass Criterion	Status	Init.

Purchase Order and delivery documentation available	Documents on file		
Instrument received undamaged – visual inspection complete	No damage		
Manufacturer's Certificate of Conformance (CoC) / Factory Calibration Certificate received	CoC on file		
Approved operating SOP (BML/SOP/QC/005) in place and accessible	SOP current		
NIST-traceable reference buffer standards procured (pH 4.00, 7.00, 9.20 ±0.01 at 25°C)	Standards available		
Environmental conditions at installation site within specification (Temp: 18–25°C; RH: 30–70%)	Within range		
Dedicated analyst assigned and trained on SOP	Training record		
Instrument registered in BML Calibration Register (BML/CAL/REG/001)	Entry made		

7. Installation Qualification (IQ)

The IQ phase provides documented evidence that the pH meter has been supplied and installed in accordance with manufacturer's specifications, approved design documents, and BML requirements.

7.1 IQ Acceptance Criteria

All IQ tests must PASS before proceeding to OQ. Any FAIL requires documented deviation and corrective action.

7.2 IQ Test 1 – Document Verification

Item	Expected / Acceptance Criterion	Result / Ref.
User Requirements Specification (URS)	URS exists and is approved prior to purchase	
Manufacturer's data sheet	Model, serial no., and specifications match PO	
Factory calibration certificate	Certificate issued within 12 months; NIST-traceable	
Operator's manual	Complete manual available in English	
Packing list / invoice	All listed items received and accounted for	

7.3 IQ Test 2 – Physical and Hardware Verification

Check	Acceptance Criterion	Observed	Pass/Fail
Main unit model	Matches PO and URS		
Serial number	Serial no. matches CoC / invoice		
Physical condition	No visible damage, cracks, or corrosion		
Display screen	Fully legible, no dead pixels		
Keypad / buttons	All functional; no sticking keys		
Electrode socket	Clean, free of damage		
ATC probe socket	Present and undamaged		
Power adaptor / cable	Correct voltage rating; cable undamaged		
Asset tag affixed	BML-EQ-PHM-001 tag visible		
Installation location	Level surface, away from vibration and direct sunlight		

Grounding	Unit properly earthed per manufacturer's instructions		

7.4 IQ Test 3 – Utilities Verification

Utility	Acceptance Criterion	Observed	Pass/Fail
Mains voltage	Within manufacturer's specified range (e.g., 220–240 V AC ±10%)		
Power socket	Dedicated, stable power supply; UPS preferred		
Environmental temperature	18–25°C at time of installation		
Relative humidity	30–70% RH		
Bench area	Non-metallic, level surface with drainage proximity		

7.5 IQ Summary and Sign-off

Role	Name	Signature	Date
Executed by (Analyst)			
Reviewed by (QAQC Head)			
Approved by (QA Manager)			

8. Operational Qualification (OQ)

The OQ phase provides documented evidence that the pH meter operates consistently within defined limits throughout its intended operational range. All tests are performed by a trained analyst following BML/SOP/QC/005.

8.1 Buffer Standards Used

Buffer	Nominal pH at 25°C	Lot / Expiry	Source	Traceability
Buffer 1	4.00 ± 0.01			NIST-traceable
Buffer 2	7.00 ± 0.01			NIST-traceable
Buffer 3	10.01 ± 0.01			NIST-traceable

8.2 OQ Test 1 – Instrument Power-On and Self-Diagnostic

Test	Acceptance Criterion	Result
Power-on sequence	Instrument powers on without error codes	
Self-diagnostic / RAM test	Internal self-test completes successfully	
Display initialisation	All display segments visible; units displayed correctly	
Date and time setting	Clock set accurately; retained after power cycle	
Electrode connection recognition	Instrument recognises connected electrode without error	
ATC probe response	ATC temperature reads ambient ± 0.5°C	

8.3 OQ Test 2 – Single-Point Calibration Verification (Neutrality)

Procedure: Rinse electrode with purified water, blot dry, immerse in pH 7.00 buffer. Allow reading to stabilise (≥1 min). Record displayed value. Acceptance: displayed pH = 7.00 ± 0.05.

Replicate	Buffer pH	Displayed pH	Deviation	Pass/Fail
1	7.00			
2	7.00			
3	7.00			

8.4 OQ Test 3 – Two-Point Calibration and Linearity (pH 4.00 and 10.01)

Procedure: Calibrate instrument using pH 4.00 buffer, then verify at pH 7.00, then pH 9.20. Record each displayed value. Acceptance criterion: ±0.05 pH units at each point.

Buffer	Reference pH	Displayed pH	Deviation	Pass/Fail
pH 4.00 buffer	4.00			
pH 7.00 buffer	7.00			
pH 10.01 buffer	9.20			

8.5 OQ Test 4 – Electrode Slope Check

The Nernst slope should be 95–105% of theoretical (59.16 mV/pH unit at 25°C). Most meters display slope% after two-point calibration. Acceptance: slope 95.0–105.0%.

Parameter	Acceptance Criterion	Observed Value
Electrode slope (%)	95.0 – 105.0%	
Asymmetry potential (mV)	≤ ±30 mV from zero	

8.6 OQ Test 5 – Temperature Compensation Verification

Procedure: Prepare pH 7.00 buffer at two temperatures (approx. 20°C and 37°C) using a calibrated thermometer. Record displayed pH and temperature at each condition. Acceptance: ATC-corrected pH within ±0.05 of certified buffer value at each temperature.

Buffer	Target Temp (°C)	Actual Temp (°C)	Expected pH	Displayed pH	Pass/Fail
pH 7.00	20		7.00 ± 0.05		
pH 7.00	37		7.00 ± 0.05		

8.7 OQ Test 6 – Response Time

Procedure: Transfer electrode from pH 7.00 buffer to pH 4.00 buffer. Record time from immersion to stable reading (drift ≤ 0.01 pH/min). Acceptance: stable reading achieved within 60 seconds.

Test	From Buffer	Time to Stable Reading (s)	Pass/Fail
Response time	pH 7.00 → pH 4.00		
Response time (reverse)	pH 4.00 → pH 10.01		

8.8 OQ Test 7 – Reproducibility (Repeatability)

Procedure: Measure pH 7.00 buffer 10 consecutive times without re-calibrating. Record all values. Acceptance: RSD ≤ 0.3% and all readings within ±0.05 pH of reference.

Reading No.	pH Displayed	Reading No.	pH Displayed	Criterion
1		6		
2		7		RSD ≤ 0.3%
3		8		
4		9		All ± 0.05 pH
5		10		
Mean ± SD:		Mean: _____ ± _____ SD; RSD: _____%		

8.9 OQ Summary and Sign-off

Role	Name	Signature	Date
Executed by (Analyst)			
Reviewed by (QAQC Head)			
Approved by (QA Manager)			

9. Performance Qualification (PQ)

The PQ phase provides documented evidence that the pH meter consistently performs within defined specifications when used for its intended purpose: measurement of pH in LVP pharmaceutical solutions. PQ is conducted under routine conditions including normal operator changeover.

9.1 PQ Test 1 – Inter-Day Precision (3 Days)

Procedure: Measure pH 7.00 buffer on three separate days (different analysts may be used). Calibrate fresh each day using pH 4.00 and 7.00 buffers. Acceptance: mean pH on each day within ±0.05; day-to-day RSD ≤ 0.5%.

Day	Analyst	Buffer pH	Displayed pH	Deviation	Pass/Fail
Day 1		7.00			
Day 2		7.00			
Day 3		7.00			

9.2 PQ Test 2 – pH Measurement of LVP Products

Procedure: Obtain a representative sample of each LVP product (as approved batch or spiked reference). Measure pH using calibrated instrument. Compare against USP/Ph.Eur. specification. Conduct in triplicate on each product.

Product	pH Specification (USP)	Result 1	Result 2	Result 3	Pass/Fail
Normal Saline 0.9% (NaCl)	4.5 – 7.0				
Dextrose 5% (D5W)	3.2 – 6.5				
Ringer's Lactate	5.0 – 7.0				

9.3 PQ Test 3 – Accuracy Against External Reference

Procedure: A certified external reference pH measurement (NIST-traceable or by accredited external lab) is compared against BML instrument readings on the same sample. Acceptance: BML result within ±0.05 pH of certified reference.

Sample	Reference pH (Certified)	BML Measured pH	Difference	Pass/Fail
Sample A				
Sample B				

9.4 PQ Test 4 – Operator-to-Operator Variability

Procedure: Two trained analysts independently measure the same pH 7.00 buffer sample (five readings each, same calibration). Acceptance: combined RSD ≤ 0.5%; mean difference between analysts ≤ 0.05 pH.

Analyst A	pH Reading	Analyst B	pH Reading	Acceptance
Reading 1		Reading 1		RSD ≤ 0.5%
Reading 2		Reading 2		
Reading 3		Reading 3		Mean diff ≤ 0.05 pH
Reading 4		Reading 4		
Reading 5		Reading 5		

9.5 PQ Summary and Sign-off

Role	Name	Signature	Date
Executed by (Analyst)			
Reviewed by (QAQC Head)			
Approved by (QA Manager)			

10. Deviation Management

Any test result that fails to meet the acceptance criterion constitutes a deviation. The following procedure applies:

- The deviation shall be documented in the Qualification Report using the Deviation Log (Section 5 of the Report).
- An immediate investigation shall be conducted to determine root cause.
- Corrective and Preventive Action (CAPA) shall be assigned with a responsible person and target date.
- The failed test shall be repeated after CAPA implementation with documented justification.
- Major deviations affecting product safety or regulatory compliance shall be escalated to the QA Manager immediately.
- A requalification decision shall be made by the QA Manager based on the severity of the deviation.

No qualification phase shall be considered complete while open deviations with incomplete CAPA remain.

11. Requalification Criteria

Full or partial requalification shall be triggered upon occurrence of any of the following events:

Trigger Event	Qualification Phase	Frequency
Routine periodic requalification	PQ (abbreviated)	Annual
Electrode replacement	OQ + PQ	As needed
Relocation to new bench or room	IQ + OQ + PQ	As needed
Major repair or firmware/software update	OQ + PQ	As needed
OOS trend identified during routine calibration	OQ + PQ	As needed
Failure identified during internal audit or inspection	Full IQ/OQ/PQ	As needed

12. Document Control

Version	Date	Description of Change	Approved by
01	April 2025	Initial issue	QA Manager

13. Appendix

Appendix A – Buffer pH vs Temperature Reference Table

Reference values per NIST SRM 185j and USP <791>. ATC should correct for these shifts automatically.

Temperature (°C)	pH 4.00 Buffer	pH 7.00 Buffer	pH 10.01 Buffer
15	4.00	7.10	10.12
20	4.00	7.07	10.06
25	4.00	7.00	10.01
30	4.01	6.99	9.97
37	4.02	6.98	9.91

Appendix B – Glossary

Term	Definition
ATC	Automatic Temperature Compensation – electronic correction of pH reading for sample temperature
IQ	Installation Qualification – verification that equipment is installed correctly
Nernst Slope	Theoretical voltage change per pH unit at 25°C: 59.16 mV/pH. Electrode slope expressed as % of theoretical
NIST	National Institute of Standards and Technology (USA) – provides traceable reference standards
OQ	Operational Qualification – verification that equipment operates within specified ranges
PQ	Performance Qualification – verification that equipment consistently performs for its intended use
RSD	Relative Standard Deviation – SD / Mean × 100%; measure of precision
URS	User Requirements Specification – document defining what the instrument must do